SJ INNOVATION · ASSESSMENT 3 (CONTINUATION OF ASSESSMENT 2)

EduHub — Level 3: integrated, observable, production-grade AI

Learning Management System · real integrations, multi-agent AI, and honest evaluation

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Where you got to in Assessment 2

A working EduHub LMS with a tool-calling **Learning Assistant** (search, describe, enrol, grades, suggest) on a Gemini edge function; two automations (course-completion → certificate, and assignment reminders on a cron); a **5,040-chunk RAG** (5,000 FAQ via Ollama embeddings) with an honest 3-set noise eval; and a level classifier. Good, honest foundation to build on.

Carry-over fixes from Assessment 2 (close these as part of A3):

- Finish the accuracy run on the **full 40-row** held-out set (the Gemini quota stopped it at 32) — run it on Ollama locally to avoid the quota.
- **Stabilise RAG deployment:** remove the temporary Cloudflare-tunnel-to-local-Ollama dependency — pre-compute and store embeddings so live search always works.
- Fix the **deployed agent timeout** (live 20s vs 45s in code).
- Re-tag `assessment-3` after the final corpus, cut a GitHub Release, and **file the submission on Dev Control Tower** (missed in A2).

Assessment 3 in one line: take EduHub and make it **integrated** (real third-party services + a campaign/CRM module), **intelligent** (multi-agent with guardrails), **measured** (a real agent/RAG evaluation harness, hybrid + re-rank), and **production-grade** (CI/CD, observability, security).

Your seven tracks

1 Third-party integration (free tier) — 20 pts

Connect **≥2 real external services** relevant to EduHub, with **one outbound action** and **one inbound webhook** (verify the signature):

Resend / SendGrid (enrolment & certificate emails) and **Google Calendar** (class / assignment schedule sync) — or **HubSpot** for a student CRM. One outbound call + one inbound webhook (e.g. email delivery status).

2 Campaign / CRM module — 15 pts

A **student re-engagement** module: learner segments (new / active / inactive), an email-campaign builder, and tracking (sent / opened / re-enrolled).

Sync to HubSpot or build it custom — it must have segments, a campaign/outreach action, and tracking.

3 Multi-agent orchestration + guardrails — 20 pts

Evolve your single Learning Assistant into a **supervised multi-agent** system: a **tutor** agent (explains a lesson), an **advisor** agent (recommends courses / a learning path), and an **enrolment** agent (takes actions), under a supervisor with shared memory and guardrails (PII redaction, low-confidence refusal, structured outputs). You already have enrol / grades tools — split and supervise them.

Required: a supervisor/router over specialist agents, shared **memory**, and **guardrails** (validation, PII redaction, low-confidence refusal, structured outputs).

4 Agent & RAG evaluation harness — 15 pts

Build an agent **task-success** eval on labelled scenarios; upgrade RAG to **hybrid (vector + full-text) with re-ranking** and add **faithfulness / answer-relevance** metrics. Your A2 baseline was matching 1.00 / noisy 0.867 / **extreme 0.133** — target a clear jump on the noisy and extreme sets and report side-by-side.

Deliver numbers: agent task-success %, and RAG hit-rate / precision@k + faithfulness / answer-relevance, side-by-side with your Assessment-2 results.

5 Model improvement / fine-tune — 10 pts

Finish what A2 started: run the level classifier on the **full held-out set** (you scored only 32/40 before the quota error), report **precision / recall / F1** (target ≥0.85), and compare Ollama-local vs Gemini.

Report precision / recall / F1 (not just accuracy). Fine-tuning (LoRA / Ollama Modelfile) earns bonus.

6 CI/CD, observability & security — 15 pts

GitHub Actions running your full test suite on push; **error tracking + logs/metrics** (e.g. Sentry free); a **Supabase advisor pass** fixing RLS/security findings; plus rate-limiting / caching. Also fix the **deployed agent timeout** (live 20s vs 45s in code).

7 Architecture docs, demo & Dev CT submission — 5 pts

An architecture diagram, the evaluation report (vs A2), a short demo, an `assessment-3` tag + GitHub Release, and — this time — actually **file the submission on Dev Control Tower**.

Tooling & how-to (all free tiers)

Third-party services

- **Email:** Resend / SendGrid free tier. **Calendar:** Google Calendar API. **CRM:** HubSpot free. **Payments:** Stripe test + webhooks. **Chat:** Slack webhooks.
- Build **one outbound call** (you call them) and **one inbound webhook** (they call you — verify the signature).

Deeper AI

- **Multi-agent:** a supervisor/router that delegates to specialist agents, each with its own tools; add **memory** and **guardrails**.
- **Hybrid RAG:** vector + keyword (Postgres full-text) retrieval, then **re-rank** the merged candidates; add **citations**.
- **Evaluation:** labelled scenarios → agent task-success %; for RAG report hit-rate / precision@k + **faithfulness** and **answer-relevance**, always vs your A2 baseline.
- **Model:** report **precision / recall / F1**; optionally fine-tune (LoRA on free Colab, or an Ollama Modelfile) and compare.

Production engineering

- **CI/CD:** GitHub Actions on every push/PR. **Observability:** Sentry free + structured logs. **Security:** Supabase advisors, fix RLS, add rate-limiting & caching.

How you will be scored (Assessment 3)

Parameter	Max
Third-party integration (≥2 services, 1 outbound + 1 inbound webhook)	20
Campaign / CRM module (segments, campaigns, tracking)	15
Multi-agent orchestration + guardrails + memory	20
Agent & RAG evaluation harness (hybrid + re-rank, metrics vs A2)	15
Model improvement (F1 reported, target; or fine-tune)	10
CI/CD + observability + security (RLS advisor) pass	15
Architecture docs, demo & Dev Control Tower submission	5
Total	**100**
Bonus — live integration in production; fine-tuned / local model used by the agent	+10

What to submit on Dev Control Tower

1. Updated **live URL** + **GitHub** (tag an `assessment-3` release).
2. **Architecture diagram** showing agents, integrations and data flow.
3. **Evaluation report** — agent task-success + RAG metrics, side-by-side vs your Assessment-2 numbers.
4. **Model metrics** (precision/recall/F1; fine-tune comparison if attempted).
5. **CI run link** (green) + a note on the observability/security you added.
6. A **3–5 min demo** and updated role credentials.

> **Golden rule:** depth over breadth, and close the A2 carry-overs first. One real integration + one solid multi-agent flow + honest evaluation beats five half-built features.